Journal of Economic Interaction and Coordination (2025) 20:1021–1043
https://doi.org/10.1007/s11403-025-00447-1

REGULAR ARTICLE

Experience goods, reinforcement learning, and social networks

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Received: 31 May 2024 / Accepted: 15 April 2025 / Published online: 21 May 2025
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Abstract
This paper explores decision-making processes in experience goods markets, emphasizing how agents learn to choose in situations where value is uncertain until after consumption. The study examines agents employing individual and social learning strategies within a multi-dimensional reinforcement learning framework, particularly in scenarios of repeated choices. Agents in our model gather insights from personal experiences and through recommendations within their social networks. The simulation results highlight the benefits of combining individual and social learning. While social learning yields consistent outcomes across agents, individual learning offers the possibility of higher rewards but also greater risks. A notable finding of this research is the development of asymmetrical influence patterns in social networks. This phenomenon refers to a tendency where certain agents become disproportionately influential in guiding others' choices, leading to a centralization in how advice is sought and followed within the network. This aspect of the model sheds light on the nuances of social dynamics in decision-making processes. The study enhances our understanding of consumer behavior in markets for experience goods, providing insights into the complex interplay of individual experiences and social influences in shaping economic decisions.

1 Introduction

Economic environments often require agents to make decisions in the face of uncertainty. Choosing optimally in these circumstances poses a significant challenge, and agents may adopt various strategies to cope. In instances where the situation occurs only once, opportunities for learning from experience are minimal. However, in contexts where agents repeatedly encounter uncertainty, it is reasonable to assume that they will engage in a learning process. This process encompasses not only the strategies employed to acquire knowledge—such as

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consulting peers or using personal experience—but also the progressive refinement of decision quality. As the agent learns, they become more adept at discerning the most effective actions to take.

To make our point clearer, let's consider the example of experience goods. These goods are unique because their intrinsic value cannot be assessed until after they have been consumed. Consequently, when an agent faces the decision to purchase an experience good, there is inherent uncertainty regarding the optimal choice. In a scenario involving repeated purchases, while the consumer may opt for the familiar item, thereby knowing its expected value, the agent remains unaware of potentially better alternatives that may be available. This situation is often referred as a multi-armed bandit. However, as the details of our model will point out later, the situation described is slightly different in the way agents behave.

Therefore, there is a trade-off between choosing a known option and exploring unknown alternatives that might lead to greater utility than the familiar choice. The assumption of this work is that in such scenarios, individuals are learning not only to select the right option from a pool of alternatives but are also discerning which decision-making method yields the most utility. Consider the example of choosing a restaurant. In large enough cities, for any ranges of budget, a consumer will be faced with many alternatives, and thus price cannot be the sole decisive factor guiding choice. There is no definitive method for making the best choice under these circumstances, and people might employ various approaches: selecting the nearest, choosing the first one that appears in an internet search, or asking acquaintances for recommendations, to name a few. Then, once a restaurant is chosen, the consumer evaluates the service based on the utility received. This feedback prompts two questions: Am I satisfied enough to come back? And do I believe that the method I used to choose this restaurant was the most effective? These two questions initiate two learning processes: one about finding the best restaurant and the other about determining the most effective method to find the best restaurant.

Three foundational assumptions underpin the model presented: Interactions are without prices and without strategic games and are conducted by individuals with bounded rationality (Simon 1955).

As discussed, many economic scenarios cannot be resolved through pricing mechanisms alone, particularly when alternatives are priced similarly. This situation is common for many experience goods, such as cultural products (books, cinema tickets, records), education, healthcare, or restaurants within the same market segment.

By 'interactions without games,' we mean situations where the agents, in this context consumers, have no incentive to deceive or engage in behavior that would diminish the utility of others. This scenario aligns with many experience goods due to the inherent nature of these products. An interaction between two consumers typically involves one recommending a product to the other, based on their own experience. Consequently, there is little to no incentive to recommend anything but the best option, as the recommending agent has already consumed the good. Furthermore, in scenarios of repeated consumption, there usually exists a delay before the good is consumed again. Thus, consumers recommend restaurants because they have already patronized them and do not foresee an immediate need to do so again.

The bounded rationality of our agents is manifested in the way they make decisions and, consequently, how they learn within this model. We propose that agents utilize reinforcement learning in this environment. Reinforcement learning embodies the 'Law of Effect' (Thorndike 1927), which posits that behaviors yielding favorable outcomes are likely to be repeated in the future, whereas those with detrimental outcomes are avoided. In our scenario, agents begin with no prior knowledge about the available alternatives and gradually learn to identify the optimal behavior (in this case, a consumption choice) over time. They improve by being afforded opportunities to repeat choices, thereby exploring different alternatives. This form of learning has been extensively examined by the behaviorist school of psychology. The significance of reinforcement in non-conscious learning has been established, notably by Thorndike (1927) and Skinner (1965), for both animals and humans. Unlike conscious learning, where individuals are aware of the learning process, reinforcement learning occurs without conscious awareness that one's actions are being influenced by past reinforcements. While humans are capable of various conscious learning mechanisms, these processes are cognitively demanding, and many of our actions are still largely governed by reinforcement learning (Brenner 2006).

In the field of economics, reinforcement learning has primarily been applied in the context of game theory (Erev and Roth 1998) and to explore social dilemmas and cooperation (Macy and Flache 2002). Significant contributions that incorporate this assumption about agents' economic behaviors include the work of Cross (2008), who develops a model in which firms are depicted as learners; Arthur (1991), who uses a similar principle to devise economic agents that act like human agents'; and Kirman and Vriend (2000), who models the Marseille fish market with both buyers and sellers engaging in reinforcement learning.

As previously mentioned, our objective extends beyond merely observing the outcomes when agents are endowed with a learning mechanism to select from a range of options. We are equally interested in exploring the consequences of agents learning about their decision-making methods.

To this end, we introduce an alternative method for selection: social learning. Social learning entails situating the agents within a network that captures their social interactions.

More specifically, agents will have the opportunity to select a good based on the recommendations of direct neighbors-essentially, their acquaintances-during each iteration.

Social learning in the literature mainly involves DeGroot learning or Bayesian learning (see Golub and Sadler 2017 and Bikhchandani et al. 2021 for extensive and recent literature reviews). Both methods differ from those we propose here. DeGroot learning involves summing information from plural neighbors, and Bayesian learning updating prior beliefs, which therefore imply having an initial belief to begin with. By contrast in the scenario we study here, we define social learning as following the recommendation of one neighbor at a time.

The influence of social networks on behavior has been a subject of interest in various fields starting from the middle of the twentieth century with the work of Katz and Lazarsfeld (2017) exploring the role of influencers on voter's decisions. Since then much progress has been made on finding where these influences

take place and on quantifying them. Notably, Christakis and Fowler (2007, 2008) have shown that health behaviors were strongly related to our social networks. The way these behaviors propagate through our social networks has then been the subject of interest of Centola (2010) who highlights the complexity of this diffusion process, often requiring more than one neighbor.

Regarding the ties between consumption and social networks, we can highlight the work on path dependence and increasing returns (See Brian Arthur 1989; David 1985; Cowan 1991; Katz and Shapiro 1985) which explain how the adoption of some technologies can be closely related to the number of consumers already using the technology. This stream of literature directly involves a consumption behavior modified by other buyers, hence implying a network effect. It is nevertheless limited in the way consumers interact and are influenced by others. It does not address the role of individual position or global structure on technology adoption.

Recent advances in modeling consumer behavior have attempted to overcome these limitations by incorporating network dynamics, learning, and bounded rationality. For instance, Bischi and Tramontana (2024) propose a model where consumers adjust their choices over time based on discrepancies between expected and realized utility, while also considering peer influence. Similarly, Naimzada and Tramontana (2009) explore a dynamic consumer model with bounded rationality, where different attractors coexist due to variations in individual decision-making processes. These studies provide strong theoretical foundations for understanding how consumption choices evolve in interconnected settings rather than being fixed.

Furthermore, research by Benhabib and Day (1981) challenges the assumption that experience-based preferences always converge to a stable equilibrium, showing instead that rational choice in a stationary environment can lead to erratic, chaotic behavior. This aligns with work by Gaertner and Jungeilges (1988), who introduce a nonlinear model of interdependent consumer behavior, demonstrating how dynamic social interactions can generate unexpected fluctuations in consumption patterns.

Empirical and simulation-based approaches further support these theoretical developments. Bargigli and Pietrini (2024) employ an agent-based model to examine the interplay between conformism and distinction in shaping cyclical consumption behaviors. Their findings challenge the notion that conspicuous consumption is confined to a wealthy class, illustrating instead how broader social structures shape demand fluctuations. Along similar lines, Jungeilges and Ryazanova (2019) investigate stochastic networks of consumer interactions and highlight how peer-driven consumption can lead to transitions between different behavioral regimes. These studies emphasize that consumption is not merely an individual optimization problem but a dynamic, socially embedded process.

Otherwise, mainstream economics has generally assumed consumers with fixed tastes uncorrelated to others' consumption. Some authors have, however, noticed it might not be totally true. Leibenstein (1950) states that some activities are more desirable when shared with others. Smith (1937) and Veblen (2017) observe that wealthy agents enjoy a consumption behavior that makes them recognizable as such, and Marshall (1890) points out that agents in general do wish for a consumption behavior that makes them different than others.

We should also notice works made on conformity and fad behavior (see, for example, Banerjee 1992; Bernheim 1994) and on the evolution of patterns of consumption (Granovetter and Soong 1983). It led (Cowan et al. 1997, 2004) to develop a model of demand with interdependence among consumers where we can distinguish three groups of reference for consumers: a peer group, a contrast group, and an aspirational group. The idea that consumers make choices to differentiate or to send signals to others has been confirmed by Gasana (2009) for consumption of clothes and automobiles in the US market and by Burgiel (2017) with Polish consumers.

These theoretical insights are further supported by growing empirical evidence that network effects shape consumption decisions beyond static economic assumptions (Bailey et al. 2022; De Giorgi et al. 2020; Agarwal et al. 2021).

All of these different approaches involve one agent's decisions, behavior, or utility being affected by the actions of others.

Our model will then approach the case of repeated choice in experience markets when agents are allowed to use either their own learning mechanism, which will be modeled as reinforcement learning, or through the use of their social networks. They then also learn, again through a reinforcement mechanism, about which method yields the most utility. There are thus two layers of reinforcement learning.

Therefore, we place agents on a network structure and allow them to interact with their neighbors to make repeated choices in experience goods market. Through repeated choices and learning from experience, agents learn which products are the best to pick. Our model also allows agents to choose between individual and social learning, allowing them to explore the benefits of both approaches and learn about how to learn in general. By combining these two types of learning, our model offers a novel approach to studying consumer decision making in markets with multiple similarly priced options.

The simulations of the model described below give results about whether individual or social learning is preferable and about the naturally emerging proportion of each type of learners. We also explore the role of curiosity and trust for optimal decisions both at individual and population levels.

The remainder of this paper is structured as follows. Section 2 introduces the model, detailing how agents make repeated consumption choices under uncertainty and how reinforcement learning governs their decision-making process. We describe both individual and social learning mechanisms, as well as the way agents can dynamically adjust their learning strategies. Section 3 presents the simulation results, comparing the effectiveness of different learning approaches and examining the emergent properties of the system, including the role of social interactions and the formation of influential nodes within the network. Section 4 concludes by summarizing the main findings and discussing potential extensions.

2 The model

2.1 Agents and products

We treat a fixed population of n agents each of which is repeatedly choosing which of 10 products to consume. Consumption lasts one period so at each period every agent repeats the decision. While the 10 alternatives all cater to the same needs, they don't all produce the same payoff/amount of satisfaction. Each product is associated with a given payoff, which remains the same for the whole simulation. Payoffs vary on a discrete scale from 1 to 10, and we refer to products by the level of payoffs they reward. (So product 6 gives a payoff of 6 to the agent for example.) Tastes are homogeneous so each product delivers the same payoff for everyone in the population and everyone prefers product i to product $i-1$. Initially, agents have no information on the utility each product delivers; they only know that there are 10 products. Consequently, at the start of the simulation, agents have no way of discerning which product could give the best payoff.

2.2 Interaction

Agents of our population will be able to interact among themselves. By interaction, we mean asking or giving advice on which product to choose for the current stage of the simulation. When giving advice, the agent will simply tell which product it would have chosen for the current round. Agents always answer solicitations from other agents. An agent who receive advice always follows it.

The interaction process takes place on a network structure that we define as follows. We use an undirected graph that represents the ties between individuals. The nature of these ties captures the possibility for any of two connected agents to advise on buying a good or service. So this type of tie can involve friendship, family, or colleagues but it might be harder to exist between an employer and its employee for example.

Agents are embedded in a social network where each agent has a set of direct neighbors with whom they can interact. These neighbors represent individuals they are connected to. Interaction occurs only between directly connected agents, meaning that information or recommendations are exchanged exclusively within these local connections.

To keep a coherent interaction scheme, advice always comes from someone who is, at least for the current round, using its own experience to choose a good. We want to avoid a situation where i asks neighbor j about which alternative to choose, j answers product A, but itself asks k and, following k's advice, chooses product B. Instead, if i asks j, and j itself asks k, and k is an individual learner, then k's advice flows back to both j and i. The rare but problematic case where i asks j, who asks k, who itself asks i (creating a cycle) is resolved by forcing one among the three to become an individual learner for the round.

2.3 Learning

Agents initially have no prior knowledge of product payoffs and must learn through repeated choices. To model this process, we implement a reinforcement learning mechanism based on the Bush & Mosteller framework (1955), following the specification in Brenner (2006). This mechanism enables agents to adjust their preferences over time based on past experiences.

Each agent selects among ten products, each associated with a fixed payoff between 1 and 10. Initially, all products are considered equally likely choices. Through experience, agents update their selection probabilities, reinforcing those that yielded higher rewards while reducing the likelihood of choosing less rewarding alternatives.

The probability update follows a reinforcement rule, ensuring that choices leading to higher payoffs become more likely in future rounds. Here, $a(t)$ denotes the product chosen by the agent at time t. The update rule distinguishes between the chosen product ($a = a(t)$), for which the probability increases, and all other products ($a \neq a(t)$), for which the probability decreases. The adjustment is given by:

$$p(a, t+1) = p(a, t) + \begin{cases} \nu \cdot \Pi(a, t) \cdot (1 - p(a, t)) & \text{if } a = a(t) \\ -\nu \cdot \Pi(a, t) \cdot p(a, t) & \text{if } a \neq a(t) \end{cases}$$

where $p(a, t)$ represents the probability of selecting product a at time t, $\Pi(a, t)$ denotes the payoff received, and ν is a learning parameter that controls the speed at which agents adapt their preferences.

To ensure that $p(a, t) \in [0, 1]$ for all a and t, we explicitly constrain the values of ν and $\Pi(a, t)$. In particular, given that payoffs $\Pi(a, t)$ are bounded above by 10 in this model, we require that $\nu \cdot \Pi(a, t) \leq 1$. This implies that $\nu \leq 0.1$, which sets an upper bound on the learning rate to guarantee that the updated probabilities remain valid.

After an agent selects a product and receives its payoff, the probabilities for all available products are revised based on the utility of the chosen product $a(t)$. The probability of selecting $a(t)$ again increases,[1] while the probabilities of all other alternatives decrease proportionally. This ensures that every choice influences not only the likelihood of repeating the same decision but also the relative attractiveness of all remaining options. Since probabilities must sum to one, increasing the probability of selecting the chosen product necessarily reduces the probabilities of all other alternatives.

The parameter ν plays a critical role in the learning process. A high value of ν results in rapid adaptation but increases the risk of premature lock-in, where an agent commits to a suboptimal choice after only a few iterations. Conversely, a very low value of ν slows learning, preventing efficient exploitation of high-payoff products.

[1] Negative reinforcement could exist in a consumption context, but this model assumes that all alternatives at least partially satisfy the consumer's need. Even if a choice is unsatisfactory, the agent lacks full information about other options, making familiarity increase its likelihood of being chosen again. Testing with negative reinforcement showed only a slight slowdown in the learning process.

To illustrate this dynamic, consider an agent choosing among four products, all initially assigned equal probabilities of 0.25, each delivering a payoff of 10, and a learning rate of $\nu = 0.08$. If the agent selects one product in the first round, its probability is updated as follows:

$$p(a, t+1) = 0.25 + 0.08 \times 10 \times (1 - 0.25) = 0.85.$$

The probabilities of all other products decrease to:

$$p(a', t+1) = 0.25 - 0.08 \times 10 \times 0.25 = 0.05.$$

In a single iteration, the probability of repeating the initial choice becomes very high, and after a second selection, it increases further to 0.97. This demonstrates how large values of ν can lead to rapid convergence toward an option, potentially preventing further exploration of better alternatives. Conversely, setting ν too low results in minimal changes in probability, prolonging the exploration phase and delaying the selection of optimal choices.

This formulation differs from the standard multi-armed bandit model, where the probability of selecting an alternative is only updated when that specific alternative is chosen. Here, each choice modifies the probability of all alternatives, as the agent becomes more familiar with the selected product relative to the others. This characteristic introduces a learning inertia, where previously chosen products continue to be favored even when better options remain unexplored.

Figure 1 illustrates the impact of different values of ν on learning dynamics. When ν is too low (blue line), agents continuously switch between alternatives without settling on a high-payoff option, leading to prolonged exploration. Conversely, when ν is too high (red line), agents rapidly commit to their first rewarding choice, potentially missing better alternatives. The optimal learning process balances exploration and exploitation, avoiding both endless searching and premature lock-in.

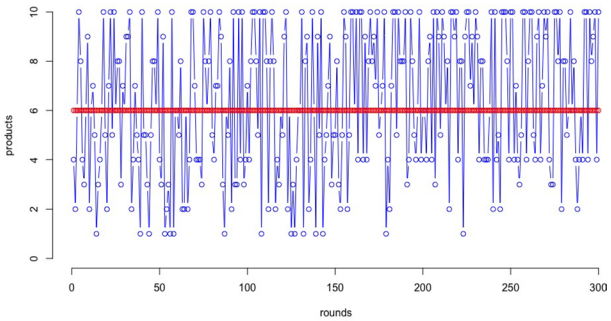


Fig. 1 Evolution of product choice for one agent other two runs. A run with a very low level of ν (in blue), and one with a very high level of ν (in red) (colour figure online)

2.4 Dynamics and learning methods

At the beginning of each period, agents are placed on a network structure that defines their set of neighbors. Each agent selects a product using one of two possible decision strategies: relying on their own experience (*individual learning*) or choosing based on a recommendation from a randomly selected neighbor (*social learning*).

Each agent i assigns a probability $p(b, t, i)$ to each behavior b, where

$$b \in B = \{\text{individual}, \text{network}\}$$

and probabilities sum to one:

$$\sum_{b \in B} p(b, t, i) = 1.$$

After making a choice, the agent receives a payoff $\Pi(a, t, i)$ from consuming the selected product. This payoff is used not only to update product selection probabilities but also to adjust the likelihood of selecting a given learning behavior in future rounds. The update rule follows a reinforcement mechanism:

$$p(b, t+1, i) = p(b, t, i) + \begin{cases} \nu_b \cdot \Pi(a, t, i) \cdot (1 - p(b, t, i)) & \text{if } b = b(t, i) \\ -\nu_b \cdot \Pi(a, t, i) \cdot p(b, t, i) & \text{if } b \neq b(t, i) \end{cases}$$

where ν_b is a learning rate parameter that controls how quickly agents adapt their choice of learning behavior. Since $\Pi(b, t, i)$ corresponds to the payoff from the selected product, we set:

$$\Pi(b, t, i) = \Pi(a, t, i).$$

This ensures that the agent updates both its *product selection probabilities* and its *learning behavior probabilities* based on the same reinforcement principle. Agents who receive high payoffs when relying on their own experience will increasingly favor individual learning, while those obtaining higher payoffs from social recommendations will shift toward network-based decision-making.

To ensure that $p(b, t, i) \in [0, 1]$ for all b, t, and i, we constrain the product of the learning rate ν_b and the payoff $\Pi(a, t, i)$. Since the payoff is bounded above by 10 in the model, we require that $\nu_b \cdot \Pi(a, t, i) \leq 1$, which implies $\nu_b \leq 0.1$. This condition guarantees that the updated behavior probabilities remain valid.

Once all agents have updated their probabilities for both product choice and learning strategy, the next period begins, and the process continues over multiple rounds.

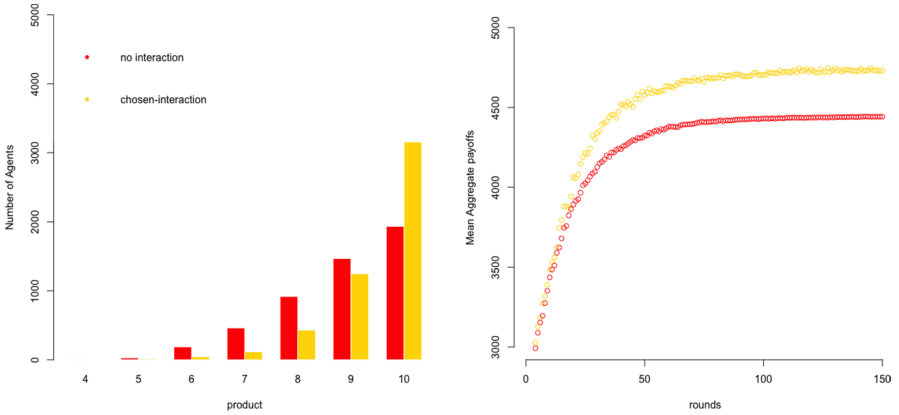


Fig. 2 Left: Distribution of product choices at last round (150) over 10 simulations of the model. Right: Evolution of mean aggregate payoffs for each round, over 10 simulations. Yellow represent results for the model with social and individual learning and red the model with only individual learning (colour figure online)

3 Results

3.1 The role of interaction

The initial outcome we examine is the contrast between two scenarios: one where agents can take their decisions based on social or individual learning and another where they operate in isolation, basing decisions solely on individual experience.

In the interactive scenario, agents are embedded within a stochastic network structured according to the Erdös–Rényi model (Erdős and Rényi 1960) with a connectivity density of 10 percent, meaning each agent has a 10 percent chance of being connected to any other agent[2] For both the interactive and non-interactive models, our simulations incorporate a total of 500 agents. The learning parameter ν is consistently set at 0.02 for both models corresponding to 20% of its maximal possible strength (Fig. 2).

In the comparative analysis, the model permitting both social and individual learning exhibits a marked advantage, as a substantial proportion of the population learns to select the highest-reward options, outperforming the model that restricts agents to individual learning. Dynamically, both models display analogous trajectories: a rapid and effective learning phase initially, which then tapers off to a slower learning rate. Despite these similarities, the interaction-enabled model achieves higher levels of aggregate payoffs from the outset, specifically within the first 20 rounds. This advantage is evident in the final round, where over $\frac{3}{5}$ of the agents successfully identify the

[2] We only display results for random network but changing the structure to ones with fat-tail degree distributions, or to ones with high cliquishness and low average path length has no effect on the dynamics observed.

optimal choice, compared to approximately $\frac{2}{5}$ in the model without interaction. Consequently, a larger fraction of the population, when interaction is not present, settles for less optimal selections.

In the model without interaction, agents initially make random product selections due to a lack of information to guide their choices. In the early stages, they explore different options and can quickly dismiss the five least rewarding products because the reinforcement received is insufficient for any one of them to become a dominant choice. However, for the five most rewarding options, the reinforcement is significant enough that some agents might start to consistently choose less optimal products like the fourth or fifth most rewarding ones, due to stochastic repetition. This phenomenon, known as the 'lock-in effect,' varies in its onset with the level of the ν parameter. With $\nu = 0.02$, the model balances exploration and exploitation: agents explore at the start but are still likely to settle on a particular choice eventually. Consequently, a significant number of agents fail to identify the most rewarding product, settling instead for suboptimal choices.

The interactive framework markedly promotes the spread of recommendations for the best products, allowing a significant number of agents to circumvent lock-in on suboptimal goods. The reason recommendations for the best products propagate first warrants a closer look. For instance, if agent i fortuitously selects A, the best restaurant in town, on their first try, the rewarding experience increases the likelihood that A will be i's recommendation to agent j in the next round. This is because A's high reward has substantially raised its standing in i's probability vector, indicating a recognition of A's quality. In contrast, if i's initial choice had been B, an average restaurant, the slight increase in B's vector probability is reflective of an unremarkable experience, making it less likely for i to recommend B again.

This creates a contamination effect, where exposure to the best goods substantially increases their promotion over others. As a result, more agents are introduced to these high-quality options, reducing the chances of settling for less. Thus, the outcomes at an aggregate level are considerably better than in models without interaction.

The reinforcement mechanism that operates within the recommendation system is crucial for enabling agents to convey information about high-quality goods. When an agent makes a recommendation to a neighbor, the suggestion is not based on their most recent selection or even the one they have chosen most frequently in the past n rounds. Instead, the recommendation reflects the choice they would likely make at that moment, which is determined by a probability vector. Thus, when an agent i is asked to recommend a good to j, it draws a good using the probabilities in its vector and gives the result to j. This vector is continually adjusted through reinforcement, ensuring that it represents the agent's up-to-date preferences. It is this dynamic aspect of the recommendation system that facilitates the spread of the most advantageous alternatives.

3.2 Social and individual learners

Once we've established that the social interaction model was performing better than the no-interaction one, we wish to understand the effects of being either a social or individual learner within the interaction model. Mainly we want to understand if one

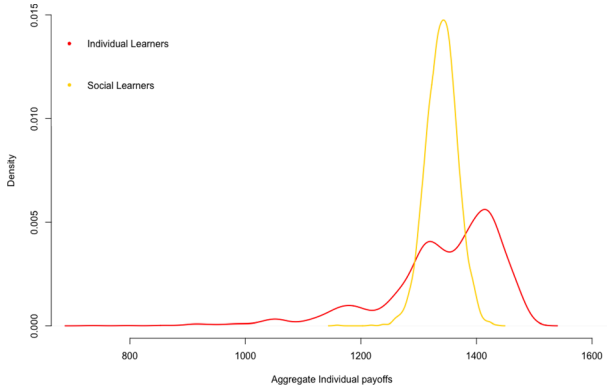


Fig. 3 Aggregate individual payoffs for the total simulation, 150 rounds in each simulation. The greatest achievable payoff being choosing product 10 at every round, yielding an aggregate payoff of 1500. In red the density for individual learners, in yellow the one for social learners. (Agents are defined between social and individual by the behavior they've adopted the most during the simulation. It yields similar results than picking the behavior they've chosen in the last rounds.) The plot displays the distributions for one representative simulation of the model

behavior is more rewarding than the other, and whether one behavior gains more popularity than the other through the simulation. To address this issue, we stick with a random network structure, inside which 500 agents are allowed to communicate among their neighborhoods. The ν parameter is still at a 20% level of its maximum potential.

Regarding the dynamics of the two behaviors, the population is split in two at the start, with half starting as individual learners and half as social learners. This is logical as agents don't have any information at the beginning and choose randomly between the two. There is then a bit of variation within the first time-steps but only to, on average (for a few hundreds of simulations), return to a 50–50% split of the population between the two types of learning. The minimum number of social learners in the final round we found was 230 and a maximum of 270. On average, agents switch learning methods 15 times during a simulation before locking-in one. The mean standard deviation is around 10, indicating a fairly high heterogeneity in switching behaviors from the agents.

There is a more interesting fact when we look at the distribution of individual aggregate payoffs over a whole simulation for both types of learners.[3] As Fig. 3 shows, the two distributions are quite different and this is confirmed with a Kolmogorov–Smirnov test ($D = 0.496,\ p < 2.2e-16$).

What we see is that potentially, being an individual learner can permit an agent to reach higher aggregate payoffs, but also lower ones than with the social learning method. For individual learners, this behavior is easily explained by the lock-in

[3] We define social and individual learners with the behavior they picked in the last round. As people lock-in in one of both early in the simulation, this is equivalent to using the behavior they have chosen the most over time.

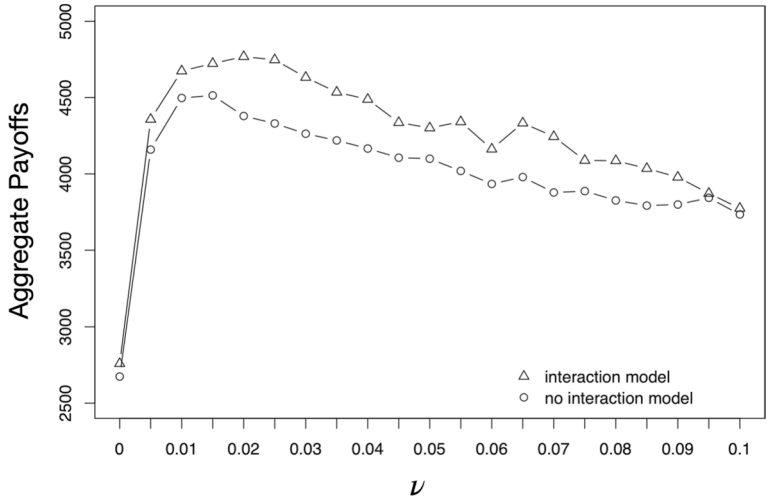


Fig. 4 Population payoff levels at last round when we vary the level of ν. Maximum possible being 5000. We do it for both the interaction and no-interaction model. $n = 500$, $Rounds = 100$

effect of the reinforcement mechanism. If you are able to find product 10 early in the simulation and lock in to it, you will be able to reach almost the highest possible level of payoffs during the simulation. But if you lock in to least-rewarding alternatives, your payoffs will be harmed for the rest of the simulation. By contrast, social learners are all reaching very similar levels of payoffs, but not as high as the best performing individual learners. This can be explained by the process of diffusion of the best alternatives that take a bit of time to propagate to all social learners. Meanwhile, social learners are sometimes being recommended low-rewarding goods.

It explains why, while the two sub-populations reach equivalent average aggregate payoffs (around 1340 for both), individual learning can be seen as a riskier behavior than social learning from an individual point of view.

3.3 Curiosity

We have so far assumed a certain level of 'curiosity' in our agents. The issue they face—navigating a pool of alternatives with unknown values—necessarily involves a trade-off between exploration and exploitation. This means agents must balance the search for better options against the choice to settle for an alternative that appears to be satisfactory.

In this framework, 'curiosity' refers to the duration dedicated to exploring unfamiliar alternatives as opposed to exploiting those already known. This includes selecting previously untried goods and reconsidering goods that didn't initially seem optimal, even after encountering better options.

To model varying degrees of curiosity, we manipulate the ν parameter. As detailed in the model description, the magnitude of ν influences the reinforcement

strength, thereby controlling the exploration phase's length before an agent commits to a specific good.

Thus, a low ν value equates to weaker reinforcement, prompting agents to explore longer before settling on a particular good. Conversely, a high ν value results in stronger reinforcement, leading to a quicker transition from exploration to exploitation of a chosen alternative.

Considering the reinforcement learning algorithm and the valuation of goods which span from 1 to 10, the upper limit for the parameter ν is set at 0.1. This represents the maximum reinforcement strength. On the opposite end of the spectrum, the minimum level of curiosity is equivalent to an absence of exploration, which would correspond to $\nu = 0$. In such a case, agents would make their choices randomly in every round, essentially eliminating the learning component of the model (Fig. 4).

The results show the clear benefits of learning and a certain level of curiosity. When we move from 0 to 0.005, there is a steep increase in payoffs, highlighting the importance of learning over random choices. The peak value of ν is relatively low, at around 0.015 for the model without interaction and 0.025 for the model with interaction. Both models exhibit similar reactions to variations of the strength parameter. Once the peak is reached, there is a gradual decline in population-level payoffs in the last round as the level of ν increases.

There is no level at which the model without interaction outperforms the interaction model.

This graph also suggests the hypothesis that the interaction model is not simply a means to 'open' social learners to new alternatives through interactions, which would imply a kind of forced' curiosity that distinguishes it from the no-interaction model. Rather, it involves the elicitation of the best alternatives that propagate through the network of agents, enabled by the reinforcement mechanism.

If interaction were merely adding a bonus level of curiosity to agents, then at the optimum level of curiosity in this model, both interaction and no-interaction models should roughly reach the same amount of payoffs, but at different levels of ν. Instead, the peak performance reached by the interaction model is never attained by the no-interaction one, clearly indicating a different mechanism at play.

3.4 Trust-based social interactions

So far, we have only considered unweighted network interactions, where a social learner randomly selects a neighbor for advice with equal probability. In our model, agents not only learn which products yield the highest payoffs but also adapt their decision-making method, choosing between individual learning and social learning. However, in the current framework, social learners do not distinguish between neighbors when seeking advice.

To extend this model, we introduce a third layer of reinforcement learning. In addition to learning about products and the choice between individual and social learning, agents now also learn which neighbors provide the best advice. This means that social learners will reinforce the probability of consulting neighbors who have previously given them high-reward recommendations. Over time, this adaptive

process could lead to a restructuring of network interactions, where certain agents become preferred sources of advice.

This additional layer of learning raises two key questions: (1) Will it improve decision-making outcomes for social learners? (2) How will the structure of network interactions evolve as some connections become more frequently utilized than others?

The process by which agents learn to identify which neighbors to consult follows the same reinforcement learning approach used for product selection and learning method selection. Specifically, after receiving advice from a neighbor j, agent i adjusts the probability of consulting j in future rounds based on the payoff received. Good advice increases this probability, while poor advice has a minimal effect, ensuring that over time, agents rely more on consistently helpful neighbors.

While agents cannot create or remove ties, this adaptive mechanism alters the *effective structure* of the network. Initially, interactions occur on an *undirected* network where all connections are equally likely to be used. Over time, as agents reinforce preferences for certain neighbors, the probability of selecting each neighbor becomes asymmetric. This effectively transforms the network into a *weighted, directed graph*, where the weight of a directed edge from agent i to agent j corresponds to the likelihood that i will seek advice from j. A weight of 0 implies that i never consults j, while a weight of 1 means i always follows j's recommendations. The nodes and ties remain the same, but when viewed through the lens of each agent's probability of asking advice from its neighbors, the network can be considered directed (since i's probability of asking j is generally different from j's probability of asking i) and weighted (as these probabilities range from 0 to 1).

This process can be interpreted as a *form of trust formation*, where reinforcement learning leads agents to increasingly favor certain sources of information. As previously discussed, reinforcement learning can result in *lock-in* to a particular alternative. In this case, the same mechanism can lead an agent to exclusively rely on a specific neighbor, even if that neighbor later changes its advice. This dynamic naturally reshapes the *flow of information* within the network, concentrating influence among certain agents.

3.4.1 Performance

We first evaluate the performance of this model regarding aggregate payoffs at last rounds, but also from an individual perspective, mainly asking if social learners, now able to choose among their neighbors, perform better than before.

Figure 5 gives the aggregate payoffs at each round, for 150 rounds, for 3 different models. We mainly want to compare a model with (in blue) and without (in red) the third layer of reinforcement. We don't see a remarkable improvement from the simpler model both regarding the payoffs levels or dynamics of learning inside populations.

What's more interesting is the evolution of density from the initial model (with 2 layers of reinforcement) and the one with trust (3 layers).

Comparison between the two models shows the distribution for social learners moves to the right, implying an improved average total payoff for a social learner

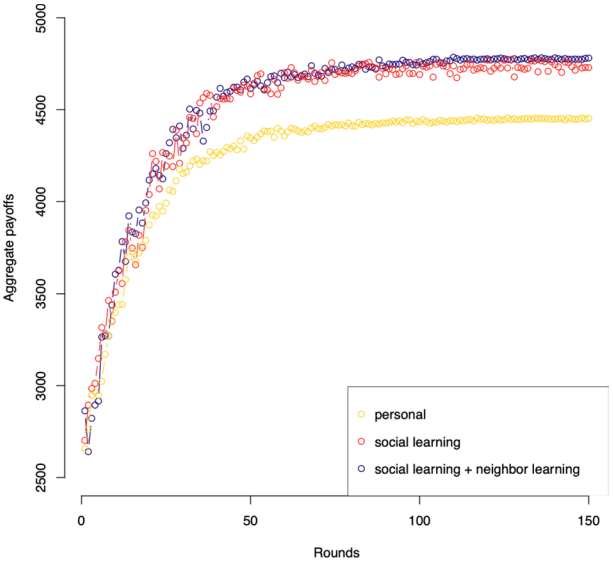


Fig. 5 At each round we plot the aggregate payoffs of the population for a representative run. We show the evolution for 150 rounds in 3 different models: When there is no social learning (in yellow), when there is social learning without weighting neighbors' ties (in red) and when we include the reinforcement of neighbors mechanism (in blue). $\nu = 0.02$, $N = 500$

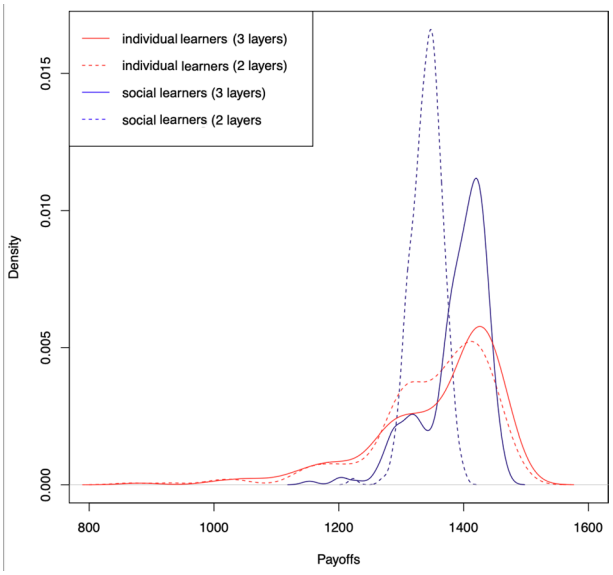


Fig. 6 We show the density of individual aggregate payoffs for 150 rounds and differentiate between agents mostly choosing to be individual learners or social learners. We compare the initial model (in dashed lines) with the one with trust (solid lines). $\nu = 0.02$, $N = 500$ (colour figure online)

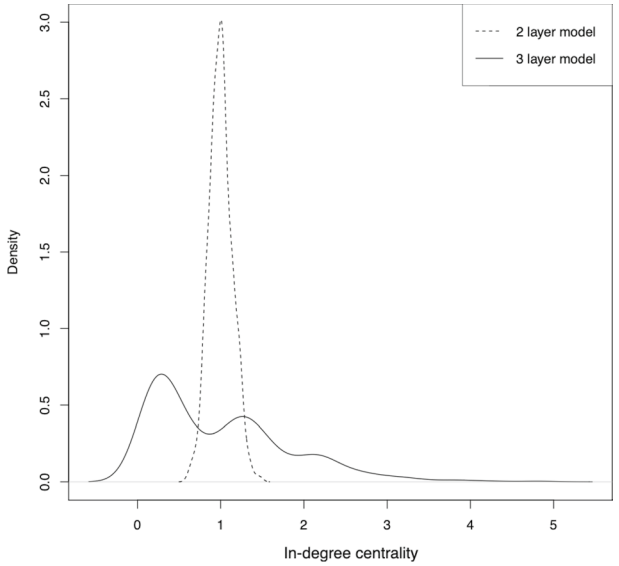


Fig. 7 We show the density of in-degree centrality for both model with two and three layers of reinforcement. $\nu = 0.02$, $n = 500$

(Fig. 6). The mean total payoff of a social learner increases by around 50 points. There would therefore be an utility for a social learner to choose a neighbor based on a reinforcement mechanism instead of choosing at random. There is, however, an increased dispersion of total individual payoffs for social learners, reflecting the same process as for individual learners: reinforcement learning can lead to lock-in on sub-optimal options, here on sub-optimal neighbors.

Regarding the proportions of each type of learners, there is only a slight increase in the proportion of social learners from the 2-layer to 3-layer model. The mean number of social learners over 10 simulations for each model only shows a 1% increase.

3.4.2 Network structure

By allowing agents to modify the probability to choose a neighbor over another, the network can be viewed as a weighted and directed one. In the initial model, the network is random which implies that the distribution of degree is Gaussian, far from real social network evidences.

We want to understand the in-degree distribution difference with the initial model. Mainly we would like to know whether some agents become more central than others. That would imply that advice is solicited from them more often than from the rest of the population.

We define in-degree centrality of an agent i as the sum of the weights that each of its neighbors j give to the tie (from j to i). Say that agent i has n neighbors. The maximum value for in-degree is n, which would imply that each neighbor always

asks advice from i rather than from their other connections. At the other extreme if the in-degree is equal to 0, all neighbors have made the choice to never ask i about which product to choose. Results from Fig. 7 show a very different distribution for the model with and without reinforcement of neighbors. While the distribution is symmetric for the two layer model, adding neighbors' reinforcement leads to a fat right tail distribution. There are therefore a few agents that will be highly seek for by their neighbors. We don't find correlation between having a higher number of neighbors on the initial random network and being central in the final one. It seems that there is a correlation between being successful and being highly central.

This might tell something interesting about the emergence of influence and opinion leaders in social networks (Fig. 8). Here, there is no preferential attachment mechanism since agents don't know the weights other agents use.

4 Conclusion

Modeling agents' behavior in the context of experience goods—where consumers learn through repeated choices without prior information—presents challenges, as individuals rely on adaptive heuristics rather than fixed optimization. Our findings suggest that reinforcement learning provides a natural framework for this process, capturing both individual and social learning dynamics. In particular, we show that social learning significantly improves aggregate outcomes, even when no agent is initially better informed. This aligns with empirical evidence from experimental economics demonstrating that individuals combine personal reinforcement learning with social information when navigating uncertainty (Fudenberg et al. 2012). In networked environments, agents often imitate successful peers, reinforcing the importance of advice structures in shaping economic behavior (Dal Bó and Fréchette 2018). These results highlight that boundedly rational agents learn through

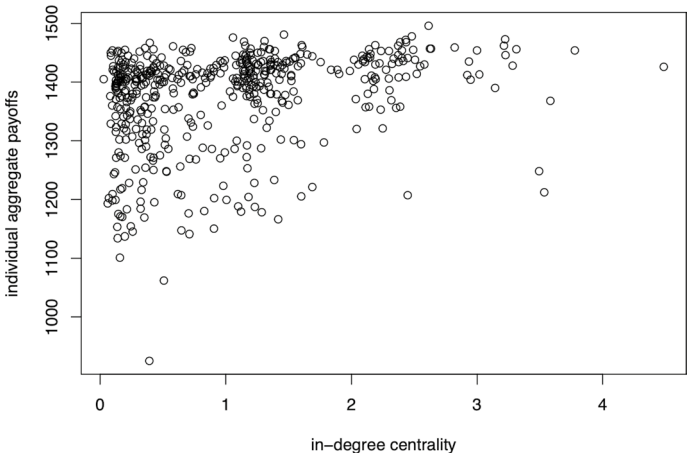


Fig. 8 Scatter plot for individual aggregate payoffs over 150 rounds and in-degree centrality at round 150. Corr 0.28, p value < 0.01. $\nu = 0.02$, $n = 500$ (colour figure online)

trial-and-error, yet do so in a way that is systematically influenced by their social context.

Beyond individual decision-making, our model also shows that social learning fosters the emergence of influential network hubs, as certain agents become disproportionately consulted for advice. This mirrors empirical findings from network economics: In organizational settings, informal advice networks tend to evolve toward structures where a few central individuals act as knowledge brokers (Aral and Van Alstyne 2011). Similarly, field studies show that economic choices such as financial adoption or insurance uptake spread more effectively when channeled through high-centrality individuals (Cai et al. 2015). These results confirm that social learning does not merely accelerate knowledge diffusion—it also reshapes the structure of economic interactions, leading to persistent centralization effects in advice-seeking networks.

This centralization, however, raises an important question: does social learning always benefit the group as a whole? While our results indicate that social learners outperform individual learners on average, this does not preclude inefficiencies in information aggregation. In our model, agents rely on reinforcement mechanisms to weigh the trustworthiness of different sources, but in real-world settings, informational cascades and echo chambers could emerge, leading to inefficient equilibria. If certain agents become central not because they provide the best advice but simply because they were consulted early on, the long-run efficiency of the system could be compromised. Similar concerns have been raised in empirical work on financial and political networks, where early movers gain disproportionate influence regardless of actual expertise (Banerjee et al. 2013). Future work could explore whether trust reinforcement in social learning systematically converges to optimal outcomes or whether suboptimal centrality effects persist even when better information exists elsewhere in the network.

Additionally, our findings contribute to broader discussions in market learning and firm behavior. While the model focuses on individual consumers, the learning mechanisms we explore are directly applicable to firms operating under uncertainty. Experimental research on adaptive pricing and Cournot competition shows that firms using reinforcement learning mechanisms-adjusting strategies based on past profits rather than full rational expectations-tend to converge to competitive market outcomes over time (Huck et al. 2004). However, the process of reaching equilibrium is shaped by the availability of social information: when firms can observe successful competitors, they tend to adapt faster and avoid prolonged inefficient learning periods (Offerman et al. 2002). This suggests that social learning mechanisms not only help individual agents but could also lead to less time elapsed before market efficiency materializes.

By integrating a third layer of reinforcement-allowing agents to learn whom to trust for advice-our model captures an essential mechanism of real-world economic learning: the development of selective trust in information sources. This process is fundamental to financial decision-making, labor market referrals, and technology adoption, where individuals rely on a mix of individual experience, social influence, and strategic trust-building. While our model assumes that agents act myopically when updating their trust in others, future research could incorporate strategic

incentives—where certain agents deliberately manipulate advice to benefit themselves, a phenomenon observed in real-world settings such as financial markets (Crawford and Sobel 1982).

More broadly, this research contributes to a growing literature on how social interactions shape economic outcomes. By formalizing reinforcement-based social learning in a network setting, we provide a framework that bridges behavioral economics, network theory, and market learning models. The results indicate that even in the absence of perfect information or rational expectations, decentralized learning processes can still lead to efficient information diffusion. However, the potential for network-driven inequalities, misallocation of influence, and path-dependent inefficiencies highlights the need for further empirical validation and refinement of these models. Future work could extend this framework by incorporating heterogeneous preferences, evolving product quality, or strategic incentives in advice-sharing-all critical dimensions of learning in economic environments shaped by both individual and collective adaptation.

Appendix 1: Simulation procedure

Model definitions

Three models are tested in this study:

- **Model 1 (No Social Learning)**: Agents individually learn which product to select using a reinforcement learning mechanism.
- **Model 2 (Individual and Social Learning)**: Agents can either learn individually or ask a randomly selected neighbor for advice.
- **Model 3 (Individual, Social, and Neighbor Learning)**: Social learners select which neighbor to ask based on past success, using a reinforcement learning mechanism.

Results are first presented in the paper for Models 1 and 2, then for Model 3.

Simulation framework

All three models follow a common simulation structure, where agents make decisions over multiple rounds. The key probabilities governing individual learning, product selection, and neighbor selection (in Model 3) evolve through a reinforcement learning mechanism based on the observed payoffs of selected products. The details of this learning process are discussed in the main paper.

Simulation steps

At each time step t, the following sequence of events occurs:

1. **Initialization**:
   - A population of n agents is created.
   - Agents are placed on an Erdös–Rényi random network with a connectivity density of 10%.
   - Each agent is assigned initial probability distributions (uniform) for:
     - Product selection.
     - Choosing between individual and social learning.
     - (In Model 3) Selecting a neighbor when using social learning.
   - The reinforcement learning parameter ν is set, along with the number of rounds.
2. **Agent Selection and Decision-Making**:
   - An agent i is randomly selected.
   - The agent chooses between **individual learning** and **social learning**:
   - The choice follows a probability distribution that evolves through reinforcement learning.
3. **If the agent chooses individual learning**:
   - The agent selects a product based on its current probability distribution.
   - The agent observes the product's payoff.
   - The agent updates:
     - The probability of selecting this product again (and thus of other products).
     - The probability of choosing individual learning in future rounds (and thus of choosing social learning).
4. **If the agent chooses social learning**:
   - **(Model 2)** The agent selects a random neighbor.
   - **(Model 3)** The agent selects a neighbor based on a probability distribution that evolves over time.
   - The neighbor recommends a product.
   - The agent picks the recommended product.
   - The agent observes the product's payoff.
   - The agent updates:
     - The probability of selecting this product again.
     - The probability of choosing social learning in future rounds.
     - **(Model 3)** The probability of asking the same neighbor again.
5. **Simulation Continuation**:
   - The process repeats until the predefined number of rounds is reached.

Author contributions All authors contributed equally to this work.

Funding The authors did not receive support from any organization for the submitted work.

Data availability Not applicable.

Declarations

Conflict of interest The authors have no conflict of interest to declare that is relevant to the content of this article.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

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